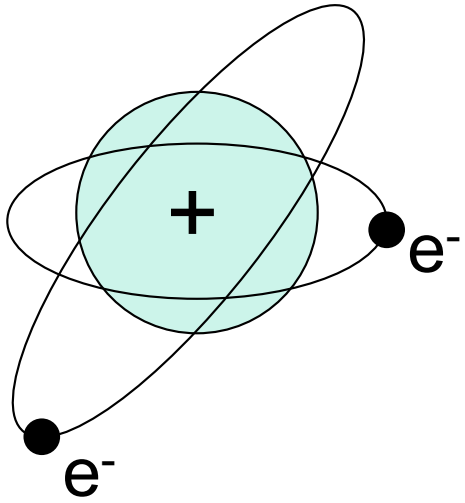
Lectures 8

Conductors

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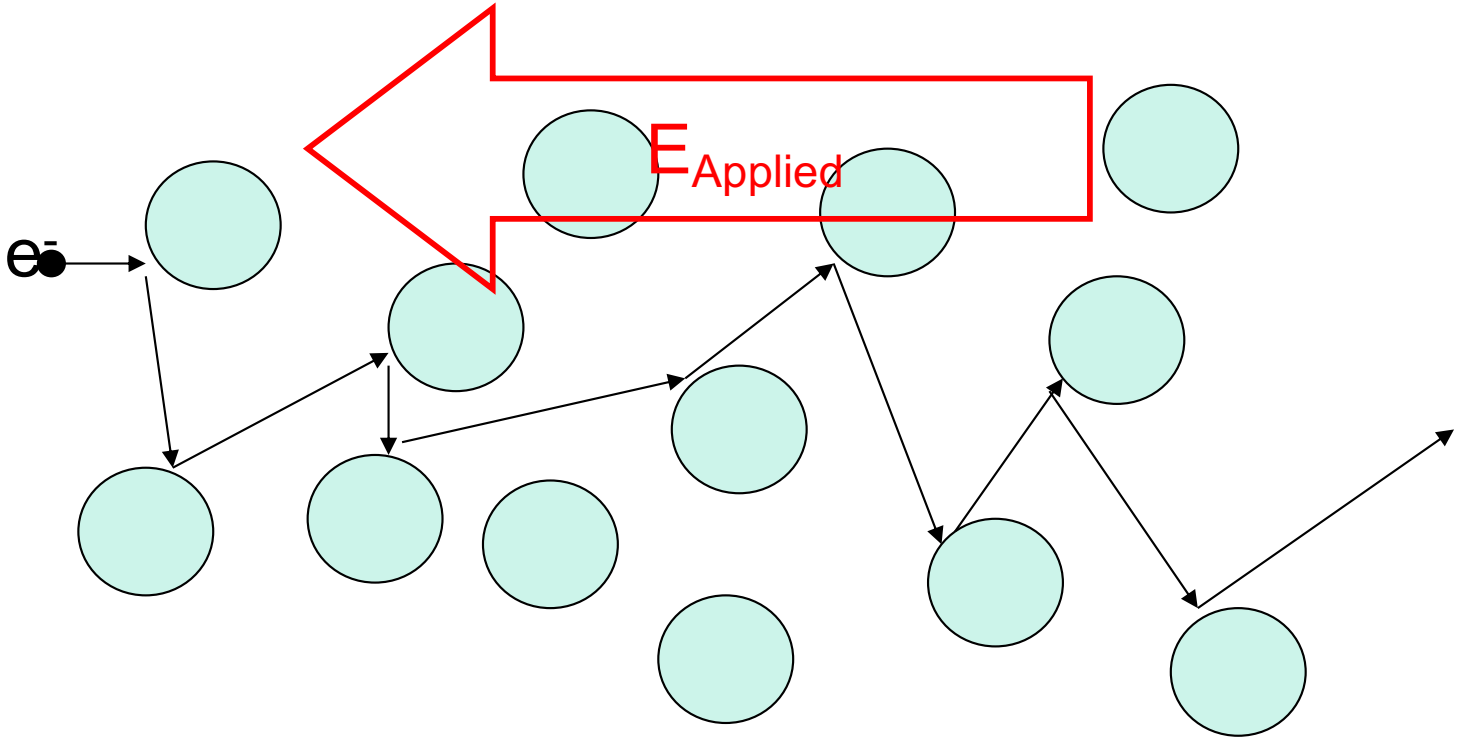
Atomic Model of Conductivity



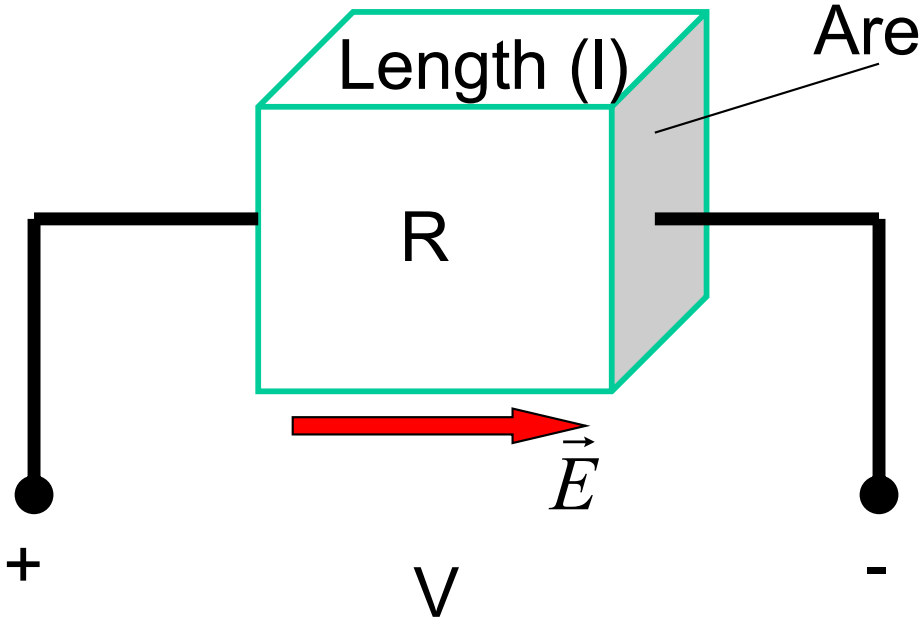
- Tightly bound inner orbitals
- Loosely bound outer orbitals
    - Free to escape the nucleus and move around inside the material

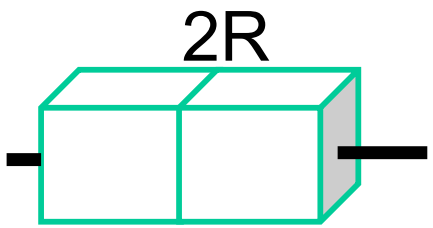
Conduction of Free Electrons

- The electron cannot travel in a straight path for very long
- It keeps running into the nuclei, getting deflected MANY times, meeting RESISTANCE

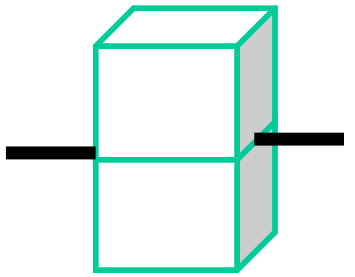


Resistors





$$\left(\frac{1}{R}+\frac{1}{R}\right)^{-1}$$



$$R = \frac{V}{I}$$

$$\Omega(Ohms) = \frac{volts}{amps}$$

R tells us how much total resistance is encountered, but does not tell us anything fundamental about the resisting material

Resistivity

$$\rho = \text{Resistivity, units of } (\Omega \cdot \text{m})$$

- Tells us how much resistance the *material* provides, factoring out the dimensions of the resistor

$$R = \rho \frac{l}{A}$$

Conductivity

$$\sigma = \frac{1}{\text{Resistivity}} = \frac{l}{R \cdot A} = \frac{1}{\Omega \cdot m}$$

- More common to describe materials in terms of conductivity rather than resistivity
- High conductivity = low resistivity
- Special units:

$$\sigma = \frac{1}{\Omega \cdot m} = \frac{Siemens}{m} = \frac{S}{m}$$

Conductivity Table

	Material	Conductivity, σ m^{-1}
Insulators	Quartz, fused	$\sim 10^{-17}$
	Ceresin wax	$\sim 10^{-17}$
	Polystyrene	$\sim 10^{-16}$
	Sulfur	$\sim 10^{-15}$
	Mica	$\sim 10^{-15}$
	Paraffin	$\sim 10^{-15}$
	Rubber, hard	$\sim 10^{-15}$
	Porcelain	$\sim 10^{-14}$
	Glass	$\sim 10^{-12}$
	Bakelite	$\sim 10^{-9}$
	Distilled water	$\sim 10^{-4}$
insulators	Dry, sandy soil	$\sim 10^{-3}$
	Marshy soil	$\sim 10^{-2}$
	Fresh water	$\sim 10^{-2}$
	Animal fat‡	4×10^{-2}
Poor conductors	Animal muscle ($\perp$ to fiber)‡	0.08
	Animal, body (ave)‡	0.2
	Animal muscle ($\parallel$ to fiber)‡	0.35
	Animal blood	0.7
	Germanium (semiconductor)	~2
Conductors	Seawater	~4
	Ferrite	10^2
	Tellurium	$\sim 5 \times 10^2$

	Material	Conductivity, σ m^{-1}
Conductors	Silicon	10^3
	Carbon	$\sim 3 \times 10^4$
	Graphite	$\sim 10^5$
	Cast iron	$\sim 10^6$
	Mercury	10^6
	Nichrome	10^6
	Stainless steel	10^6
	Constantan	2×10^6
	Silicon steel	2×10^6
	German silver	3×10^6
	Lead	5×10^6
	Tin	9×10^6
	Phosphor bronze	10^7
	Brass	1×10^7
	Zinc	1.7×10^7
	Tungsten	1.8×10^7
	Duralumin	3×10^7
	Aluminum, hard-drawn	3.5×10^7
	Gold	4.1×10^7
	Copper	5.7×10^7
	Silver	6.1×10^7
Super-conductors	Hg (at <4.1 K)	∞
	Nb (at <9.2 K)	∞
	$Nb_3(Al\text{-}Ge)$ (at <21 K)	∞
	$YBa_2Cu_3O_7$ (at <80 K)	∞

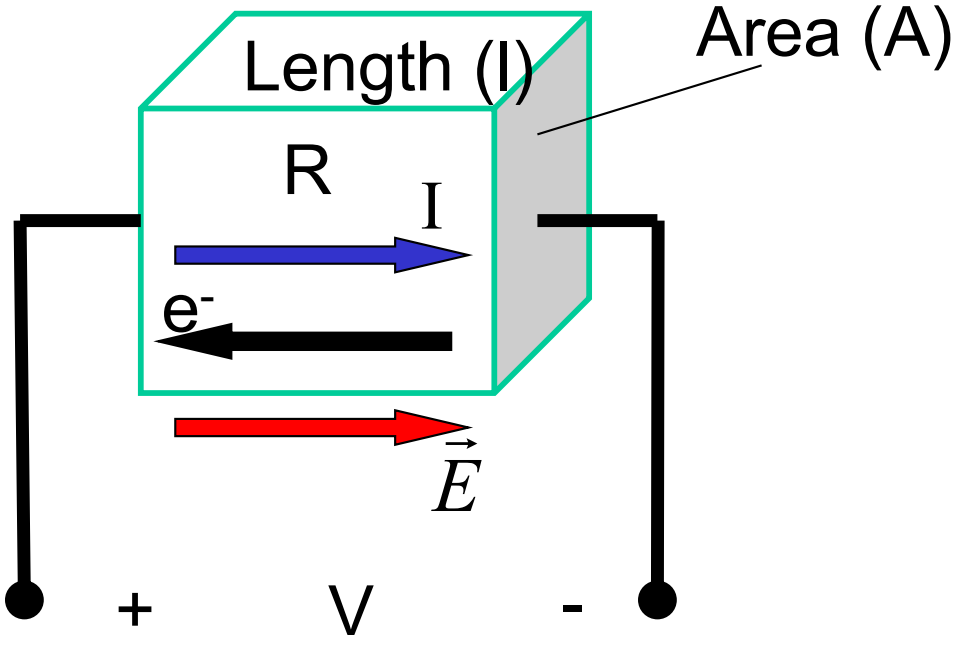
‡ or low frequencies. At 20°C except where noted.

Ohm's Law

$$|E| = \frac{V}{l} \quad \left(\frac{V}{m}\right)$$

$$|J| = \frac{I}{A} \quad \left(\frac{A}{m^2}\right)$$

$$R = \frac{l}{\sigma A} \quad (\Omega)$$



$$V = IR$$

$$E \cdot l = (J \cdot A)\left(\frac{l}{\sigma A}\right) = \frac{1}{\sigma} J \cdot l$$

$$\boxed{\vec{J} = \sigma \vec{E}}$$

Maxwell's Equations in a Conductor

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

In free space

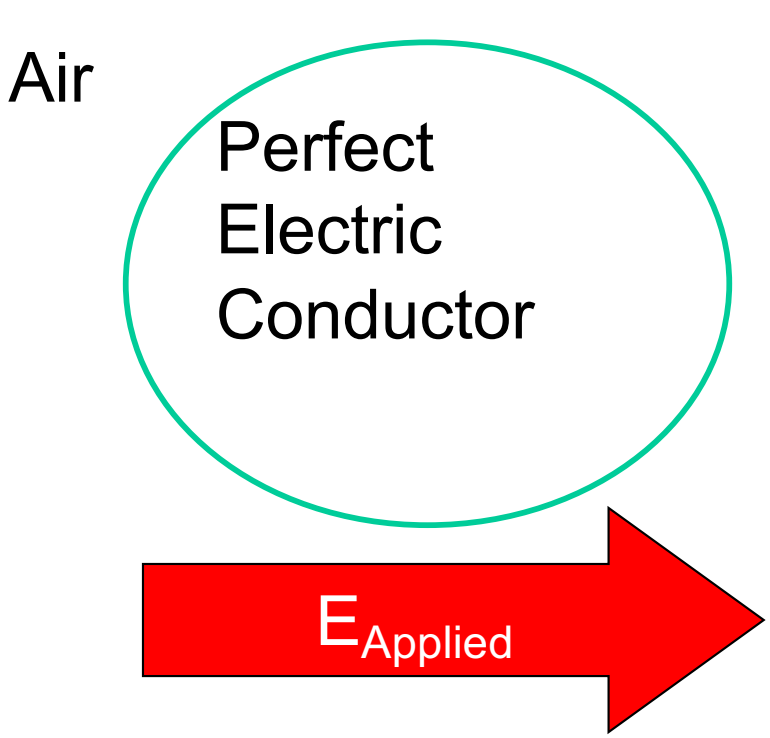
$$\vec{\nabla} \times \vec{H} = \sigma \vec{E} + \frac{\partial \vec{D}}{\partial t}$$

In a conducting material

Conduction current: Moving charge

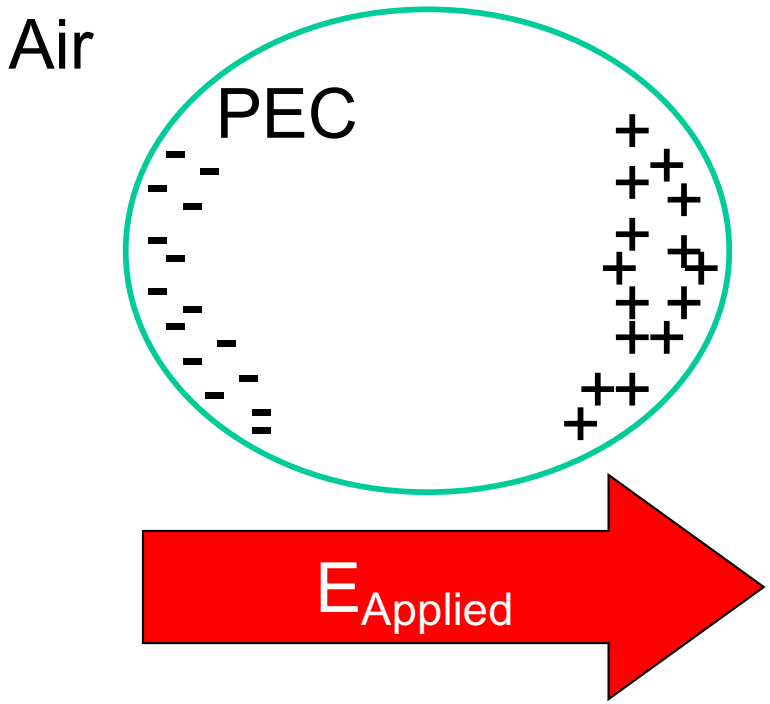
Displacement Current: Time varying E-field

Perfect Electric Conductor (PEC) in an Applied Electric Field



What Happens?

PEC in an Applied Electric Field

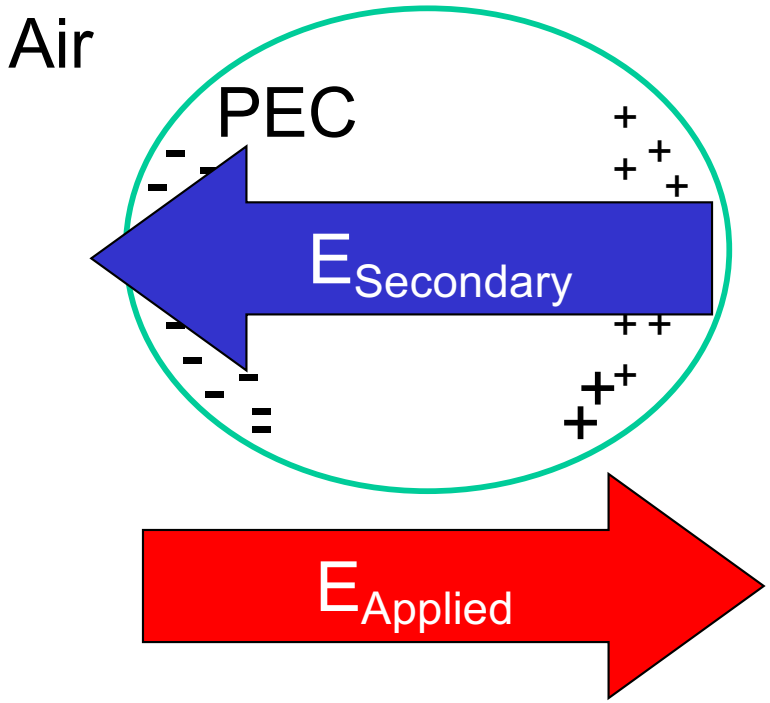


What Happens?

-Free e^- move in direction opposite of applied E

- When e^- moves away from its nucleus, it leaves a + charge behind
- The material as a whole is still NEUTRALLY CHARGED but the charge has now been redistributed

E=0 inside PEC



What Happens Next?

-The separated + and - charges create their own SECONDARY INTERNAL E field that *EXACTLY CANCELS* the applied E field

- The TOTAL E field inside the perfect conductor is ALWAYS ZERO

- If it weren't zero, free charge would continue drifting till it is!

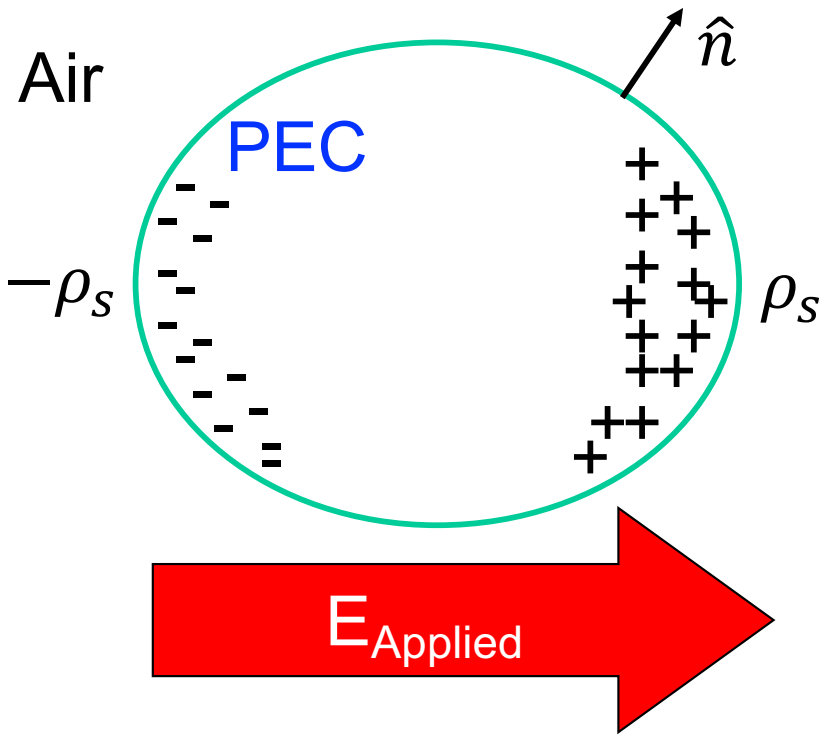
Conductors and Dielectric

- $\sigma \to \infty$ corresponds to a perfect electrical conductor (PEC)
  - $\vec{\boldsymbol{E}} = \mathbf{0}$ inside independent of $\vec{\boldsymbol{J}}$
  - in analogy with $V = 0$ across a short circuit element

$\vec{\boldsymbol{E}} = \mathbf{0}$ in PEC unconditionally

- $\sigma \to 0$ corresponds to a perfect dielectric
  - $\vec{\boldsymbol{J}} = 0$ inside independent of $\vec{\boldsymbol{E}}$
  - in analogy with $I = 0$ through an open circuit element

Review: Electrostatic field and PEC



Given $\vec{\boldsymbol{E}} = \mathbf{0}$

1. Potential V=constant on PEC surface.

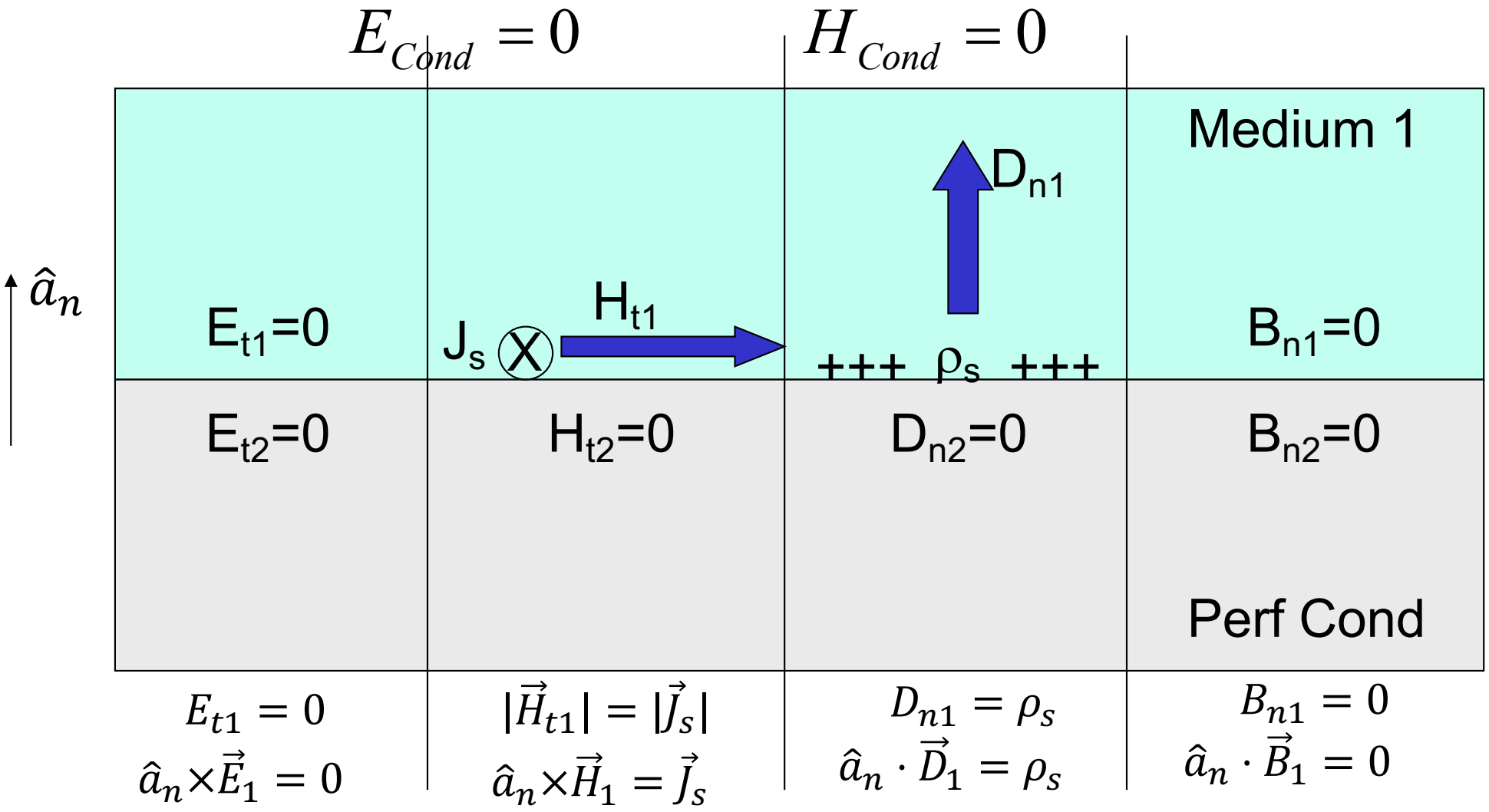
2. Gauss's Law

$$\rho = \nabla \cdot \vec{\boldsymbol{D}} = \nabla \cdot \epsilon \vec{\boldsymbol{E}} = 0$$

3. Boundary condition equations on PEC

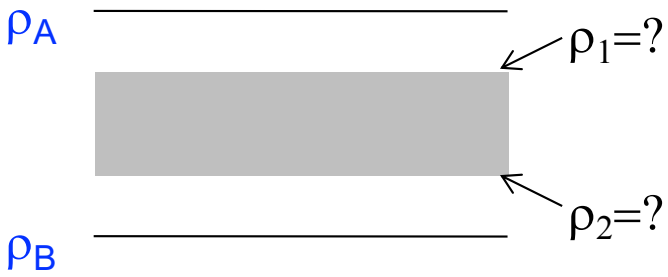
$$\hat{n} \cdot \vec{\boldsymbol{D}} = \rho_s \quad \text{and} \quad \hat{n} \times \vec{\boldsymbol{E}} = 0$$

Surface of Perfect Conductor



Infinite plane conducting slab

- An electrically neutral infinite plane conducting slab lies between two infinite plane sheets of uniform surface charge density $\rho_A > 0$ and $\rho_B > 0$. Find the surface charge densities ρ_1 and ρ_2 on the two slab surfaces.



If $\vec{E} = 0$ inside a PEC, how about $\vec{H}$?

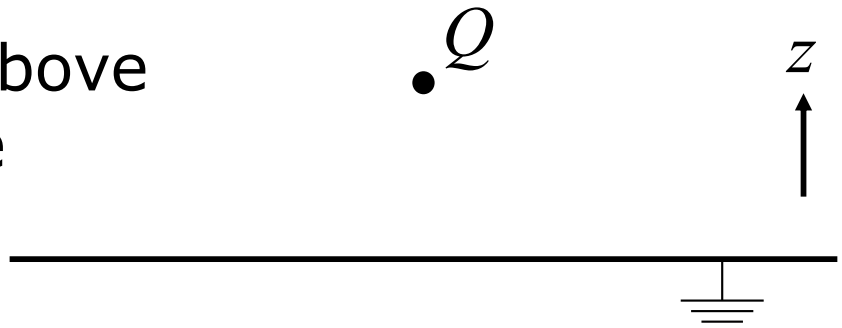
$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \Rightarrow \quad 0 = -\frac{\partial \vec{B}}{\partial t}$$

B and H must be STATIC

Technically, a non-zero static H-field can exist inside a PC, but how did it get there? It must have existed there for all time. But what created it in the first place? This is an ill-posed problem so in 329, we will assume $\vec{H} = \mathbf{0}$ inside a PEC.

Challenge Question: Point charge above a sheet

- A positive charge is located above a perfectly conducting infinite ground plane at z=0



(a) **E**=0 above the plane

(b) **E**=0 below the plane

(c) A uniform charge density is induced on the plane

(d) The total induced charge on the plane is +Q

(e) The electric field lines have a radial component at the plane

Lecture 8 Summary

- Ohm's Law: _____
- Conductivity: $\sigma = \frac{1}{\text{Resistivity}} = \frac{l}{R \cdot A}$
- Inside a perfect conductor, $\vec{\boldsymbol{E}} = ?$
- Next class
  - Dielectrics